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Increases in Lower-Body Strength Transfer Positively to Sprint Performance: A Systematic Review with Meta-Analysis

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Abstract

Background Although lower-body strength is correlated with sprint performance, whether increases in lower-body strength transfer positively to sprint performance remain unclear.

Objectives This meta-analysis determined whether increases in lower-body strength (measured with the free-weight back squat exercise) transfer positively to sprint performance, and identified the effects of various subject characteristics and resistance-training variables on the magnitude of sprint improvement.

Methods A computerized search was conducted in ADONIS, ERIC, SPORTDiscus, EBSCOhost, Google Scholar, MEDLINE and PubMed databases, and references of original studies and reviews were searched for further relevant studies. The analysis comprised 510 subjects and 85 effect sizes (ESs), nested with 26 experimental and 11 control groups and 15 studies.

Results There is a transfer between increases in lower-body strength and sprint performance as indicated by a very large significant correlation ($r = -0.77$; $p = 0.0001$) between squat strength ES and sprint ES. Additionally, the magnitude of sprint improvement is affected by the level of practice ($p = 0.03$) and body mass ($r = 0.35$; $p = 0.011$) of the subject, the frequency of resistance-training sessions per week ($r = 0.50$; $p = 0.001$) and the rest interval between sets of resistance-training exercises ($r = -0.47$; $p \leq 0.001$). Conversely, the magnitude of sprint improvement is not affected by the athlete's age ($p = 0.86$) and height ($p = 0.08$), the resistance-training methods used through the training intervention, ($p = 0.06$), average load intensity [% of 1 repetition maximum (RM)] used during the resistance-training sessions ($p = 0.34$), training program duration ($p = 0.16$), number of exercises per session ($p = 0.16$), number of sets per exercise ($p = 0.06$) and number of repetitions per set ($p = 0.48$).

Conclusions Increases in lower-body strength transfer positively to sprint performance. The magnitude of sprint improvement is affected by numerous subject characteristics and resistance-training variables, but the large difference in number of ESs available should be taken into consideration. Overall, the reported improvement in sprint performance (sprint ES = -0.87 , mean sprint improvement = 3.11 %) resulting from resistance training is of practical relevance for coaches and athletes in sport activities requiring high levels of speed.

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Key Points

There is a transfer of lower-body strength training to sprint performance

The magnitude of sprint improvement is affected by the level of practice and body mass of the subject, the frequency of resistance-training sessions per week and the rest interval between sets of resistance-training exercises

The improvement in sprint performance resulting from resistance training is of practical relevance for coaches and athletes in sport activities requiring a high level of speed, especially over short/medium distances (<30 m)

1 Introduction

Enhancing sprint performance is a fundamental component of training interventions designed to stimulate the improvements required for success in many individual and team sports. Indeed, sprint performance has been shown to be a major determinant in accessing a higher level of performance capacity in soccer [1], American Football [2] and rugby league [3], while also playing a large role in dictating selection to a starting position on many teams [2–4]. Considerable literature reports a large to very large relationship between lower-body strength measured with the back squat exercise, and sprint performance, suggesting that increasing lower-body strength is fundamental when attempting to improve sprint performance [5–9]. For example, Seitz et al. [8] found a significant correlation ($r = -0.57$, $p = 0.04$) between back squat strength and 20-m sprint time among junior elite rugby league players. Additionally, a similar relationship ($r = -0.66$, $p < 0.05$) exists between back squat strength and 40-m sprint time among professional rugby league players [5].

The strong relationship between back squat strength and sprint performance might be explained by the fact that individuals exhibiting greater lower-body strength are able to produce a higher peak ground reaction force (pGRF), impulse, and rate of force development (RFD) during each foot strike while running. It is clear from the scientific literature that an individual's overall sprint performance or ability to express higher sprint velocities is impacted by his ability to express high pGRF, and impulse [10]. However, whether resistance-training-induced increases in lower-body strength transfer positively to sprint performance remains unclear. While previous studies report concurrent improvements in lower-body strength and sprint

performance following a resistance-training intervention [11–13], other studies have failed to demonstrate faster sprint times in relation to increases in lower-body strength [14–16]. A possible explanation for this discrepancy might be that the effect of a resistance-training intervention on athletic performance is influenced by several subject characteristics, such as the level of practice [17] or chronological age [18]. Likewise, various resistance-training variables, including the duration, volume, intensity and methodology of training, can also influence the transferability of training-induced strength gains to the targeted athletic (sprint) performance. Therefore, there is no clear agreement in the scientific literature, regarding the optimal combination of these variables to achieve maximum gains in sprint performance. By using meta-analytic techniques, it may be possible to gain a greater understanding about the effect of resistance-training-induced increases in lower-body strength on sprint performance improvement. Additionally, such research may provide a precise estimate of which resistance-training variables best contribute to improving sprint performance.

1.1 Objectives

The purpose of this systematic review with meta-analysis was to (1) determine whether increases in lower-body strength positively transfer to sprint performance and (2) establish the relative importance of various subject characteristics and resistance-training variables on sprint improvement. The central hypotheses of this investigation are that increases in lower-body strength would lead to greater improvements in sprint performance and several subject characteristics and training variables would affect the magnitude of sprint improvement.

2 Methods

2.1 Literature Search

A search was performed using the following keywords in the English, French and Spanish languages: 'strength training', 'sprint training', 'squat training', 'sprint performance', 'sprint times', 'velocity', 'entraînement force', 'entraînement squat', 'entraînement vitesse', 'entraînement sprint', 'fuerza', 'velocidad'. These keywords were applied in the databases ADONIS, ERIC, SPORTDiscus, EBSCOhost, Google Scholar, MEDLINE and PubMed. Additionally, the reference lists and citations of the identified studies were explored using Google Scholar to find additional articles. Attempts were also made to contact the authors of the selected articles to request any missing relevant information. The present meta-analysis includes

studies that (1) have presented original research data on healthy human subjects and (2) are published in peer-reviewed journals. No age, sex or language restrictions were imposed during the search stage.

2.2 Inclusion and Exclusion Criteria

Research studies implementing resistance-training programs for lower-limb muscles were the primary focus of the literature search. Studies implementing training programs for both lower- and upper-limb muscles were also accepted. Conversely, studies that examined only the training of the upper-limb musculature were excluded from this meta-analysis. A total of 171 studies were initially identified for further scrutiny.

The next step was to select studies with respect to their internal validity. Selection was based on the recommendations by Campbell and Stanley [19] and included (1) randomized control studies, (2) studies using instruments with high reliability and validity, (3) studies where the sprint test was conducted pre- and post-training and (4) studies where the strength test was conducted using a free-weight (full, parallel or half) back squat exercise. After critically analyzing the initial studies collected with the above criteria, a cohort of 15 studies was selected (Fig. 1 [20–34]).

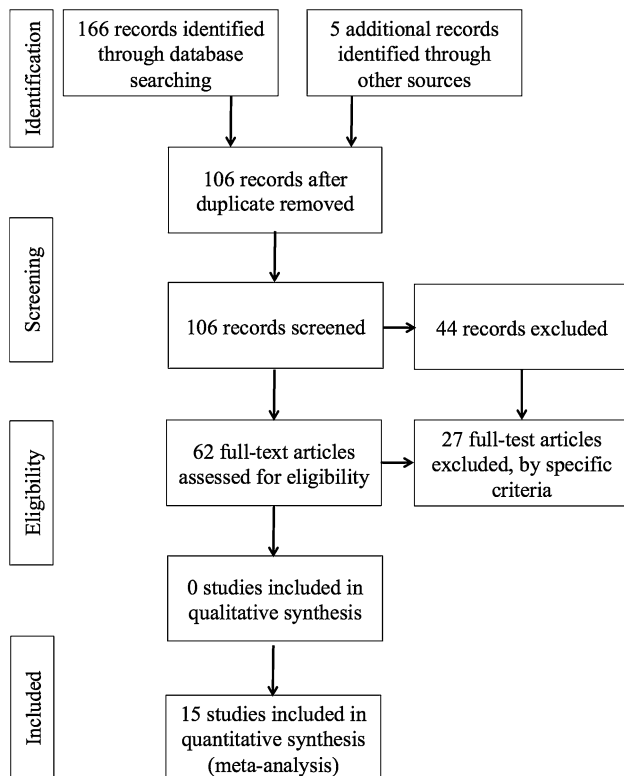


Fig. 1 Flow diagram of the studies that underwent the review process

2.3 Data Extraction and Quality Assessment

Each study was then read and coded by two independent investigators using different moderator variables. Because training efficiency can be affected by several variables, independent variables were grouped into the following categories: (1) subject characteristics: body weight, height, age and level of practice; (2) resistance-training program elements: back squat training method, loaded jump squat/ countermovement jump (loaded JS/CMJ) training method, combination of back squat, plyometric and loaded JS/CMJ training method, average load intensity [% 1 repetition maximum (RM)], frequency of sessions per week, program duration, average number of exercises per session, average number of sets per exercise, average number of repetitions per set, and average rest intervals between sets of exercises; and (3) outcome measurements: the distance of the sprint test(s) used to assess sprint performance. The mean agreement was calculated by an intra-class correlation coefficient (ICC). For such coding methods, a mean agreement of 0.90 is generally accepted as an appropriate level of reliability [35]. A mean agreement of 0.93 was calculated in the present investigation, which is well above the 0.90 mark for acceptable reliability. The investigators examined and resolved any coding differences before the final analysis.

2.4 Analysis and Interpretation of Results

The effect size (ES) is a standardized value that allows the determination of the magnitude of the differences between groups or experimental conditions [36]. The ESs were calculated using Hedges and Olkin's g [35], using the following formula [1]:

$$g = \frac{(M_{\text{post}} - M_{\text{pre}})}{SD_{\text{pooled}}}$$

where M_{post} is the mean of the post-sprint test, M_{pre} is the mean of the pre-sprint test, and SD_{pooled} is the pooled standard deviation of the measurements [2]:

$$SD_{\text{pooled}} = \frac{((n_1 - 1) \times SD_1^2 + (n_2 - 1) \times SD_2^2)}{(n_1 + n_2 - 2)}$$

where SD_1^2 is the standard deviation of the pre-sprint test and SD_2^2 is the standard deviation of the post-sprint test.

It has been suggested that the ES should be corrected for the magnitude of the sample size of each study because the absolute value of the ES is overestimated in small sample sizes [35–37]. Therefore, a correction factor was calculated using the following formula [35]:

$$\text{Correction factor} = 1 - \frac{3}{4(n_1 + n_2 - 2) - 1}$$

The corrected ES was calculated using the following formula:

$$\text{Corrected ES} = g \times \text{correction factor.}$$

An analysis of variance (ANOVA) was used to examine the effect of categorical independent variables (i.e., group, level of practice, training level, sport activity, resistance-training methods, average load intensity and distance of the sprint test) on sprint ES, [36–38]. In the case of quantitative independent variables (i.e., age, body weight, height, frequency of resistance-training sessions per week, training program duration, number of exercises per session, number of sets per exercise, number of repetitions per set, rest intervals between sets of exercises) a Pearson's (r) correlation test was used to examine the relationships between the sprint ES and the variable values [37]. Statistical significance was set at $p \leq 0.05$ for all analyses. The scale used for interpretation was specific to training research and based upon the one proposed by Hopkins [39] to evaluate the relative magnitude of an ES. The magnitude of the ESs was considered trivial (<0.2), small (0.2 – 0.59), moderate (0.60 – 1.19), large (1.2 – 1.99) or very large (>2). Strength of relationships was assessed using the following criteria [40]: trivial ($r < 0.1$), small ($r = 0.1$ – 0.3), moderate ($r = 0.3$ – 0.5), large ($r = 0.5$ – 0.7), very large ($r = 0.7$ – 0.9) and nearly perfect ($r > 0.9$). An Egger's test was developed to address the potential of publication bias relating to small-study sample size.

3 Results

The Egger's test showed no small-study effect ($p = 0.161$). There was a very large statistically significant correlation between squat ES and sprint ES [$r = -0.77$; $r^2 = 0.60$; $p \leq 0.001$; 95 % confidence interval (CI) -0.85 to -0.67], suggesting that increases in lower-body strength transferred positively to sprint performance (i.e., decrease in sprint time) (Fig. 2). Additionally, there was a statistically significant difference ($p < 0.001$) between the different groups, with the experimental groups (i.e., undertaking resistance training) displaying a greater sprint ES (ES = -0.87) in comparison to the control groups (ES = 0.02) (Table 1). A forest plot depicting the sprint ESs and associated 95 % CI is shown in Fig. 3.

With respect to the subject characteristics, there was a statistically significant moderate relationship ($r = 0.35$; $p = 0.011$) between body mass, and sprint ES. Additionally, there was a statistically significant difference ($p = 0.03$) between the different levels of practice of the subjects, with national athletes displaying a greater sprint ES in comparison to the other subjects. Conversely, the

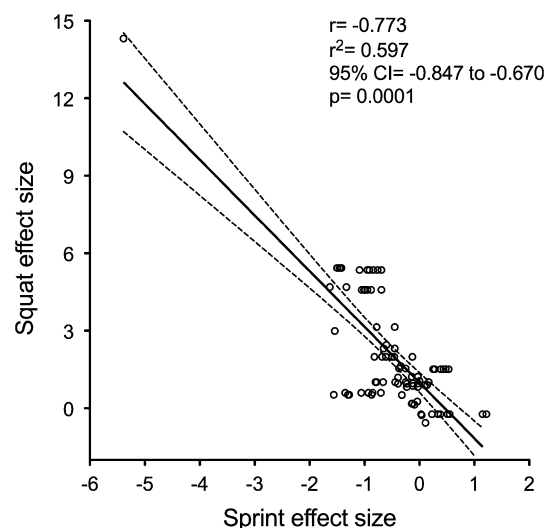


Fig. 2 Correlation between squat and sprint effect sizes ($n = 85$). The two dashed lines represent the 95 % CI band. CI confidence interval, p p value, r Pearson's correlation coefficient

correlations between age ($r = 0.03$; $p = 0.86$), as well as height ($r = 0.26$; $p = 0.08$), and sprint ES were not statistically significant.

With respect to the resistance-training program elements, there was no statistically significant difference between the different resistance-training methods used during the training interventions ($p = 0.06$; ESs = -0.29 to -1.20) (Table 2). Similarly, no statistically significant differences ($p = 0.34$) were found among the different average intensities (the average % 1 RM) used throughout each resistance-training intervention.

There was a moderate statistically significant relationship between the frequency of training sessions performed per week ($r = 0.50$; $p = 0.001$) and the average rest interval between sets ($r = -0.47$; $p \leq 0.001$), and sprint ES. Conversely, the program duration ($r = -0.20$; $p = 0.16$), average number of exercises per session ($r = -0.20$; $p = 0.16$), average number of sets per exercise ($r = -0.27$; $p = 0.06$) and average number of repetitions per set ($r = -0.10$; $p = 0.48$) were not correlated with sprint ES (Table 3).

With respect to the test outcome, there was no statistically significant difference in sprint ES ($p = 0.24$) among the different sprint test distances (Table 4).

4 Discussion

The purpose of this meta-analysis was to (1) determine whether increases in lower-body strength (measured with a full, parallel or half back squat exercise) transfer positively to sprint performance and (2) examine the effects of various subject characteristics and training variables on the

Table 1 Analysis for independent variables of subject characteristics

Subject characteristics								
Independent variables	Sprint improvement (%), mean $\pm$ SD	<i>F</i>	<i>p</i>	ES	SD	<i>n</i>	<i>r</i>	<i>p</i>
Age						52	0.03	0.86
Body mass						52	0.35	0.011
Height						48	0.26	0.08
Group		<i>F</i> (1,83) = 30.68	<0.001					
Experimental	3.11 $\pm$ 2.27			−0.87	0.81	52		
Control	0.05 $\pm$ 2.13			0.02	0.52	33		
Level of practice		<i>F</i> (3,46) = 2.96	=0.03 ^a					
International	2.34 $\pm$ 0.83			−0.53	0.18	5		
National	4.35 $\pm$ 2.05			−1.24	0.99	22		
Regional	0.97 $\pm$ 0.13			−0.31	0.08	4		
Others	2.44 $\pm$ 2.36			−0.67	0.62	19		
NR	1.96 $\pm$ 2.12			−0.52	0.56	2		

ES effect size, *F* Fisher–Snedecor distribution, *n* sample size, NR not reported, *p* *p* value, SD standard deviation, *r* Pearson's coefficient correlation

^a NR group was not included in the analysis of variance

magnitude of sprint performance improvement. Our hypotheses are supported by the present data, as increases in lower-body strength transfer positively to sprint performance (i.e., decrease in sprint time) and the magnitude of sprint improvement is influenced by several subject characteristics and training variables.

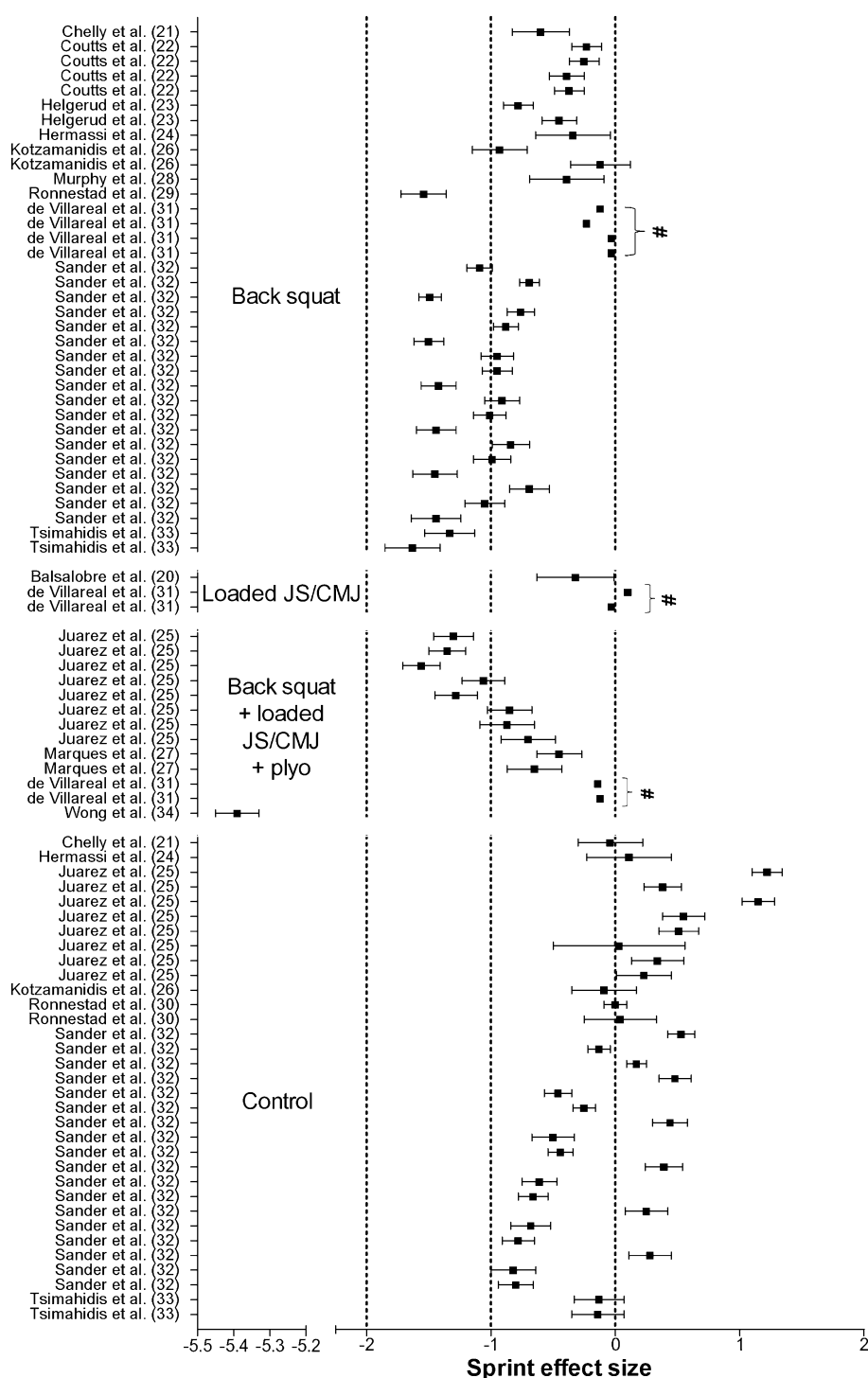
4.1 Transfer of Lower-Body Strength to Sprint Performance

Although there is a large volume of published studies reporting a significant correlation between lower-body strength and sprint performance [5–7, 9, 41], whether there is a transfer between increases in lower-body strength and sprint performance remained unclear. Previous studies reported concurrent increases in back squat strength and sprint performance after a resistance-training intervention [11–13], while others failed to demonstrate that increases in back squat strength resulted in a parallel improvement in sprint performance [14–16]. For example, Comfort et al. [11] observed a concurrent increase in mean back squat strength (+17.7 %) and decrease in mean sprint time over 5 (−7.6 %), 10 (−7.3 %) and 20 m (−5.9 %) following an 8-week resistance-training intervention in professional rugby league players (1 RM squat kg per kg body mass = 1.78). Similarly, Harris et al. [15] demonstrated that an 11.64 % increase in mean back squat strength resulting from a 9-week resistance-training intervention positively transferred to 27.43-m (30-yard) (−1.36 %) mean sprint time in university football players (1 RM squat kg per kg body mass $\geq$ 1.40). Conversely, smaller increases in mean back squat strength (+3.62 and +9.85 %) failed to

positively transfer to 27.43-m (30-yard) (+0.78 and 0 %, respectively) mean sprint times [20]. These findings suggest that the greater the improvement in back squat strength, the greater the improvement in sprint performance. In the present meta-analysis, including 15 studies and 85 groups of subjects (total number of subjects 510), the very large significant correlation ($r = -0.77$; $p \leq 0.001$; 95 % CI −0.85 to −0.67) between squat strength ES and sprint ES (decrease in sprint time) strongly suggests that increases in lower-body strength positively transfer to sprint performance (Fig. 2). This finding appears logical since it has been shown that pGRF, impulse and RFD during each foot strike while running significantly impact the athlete's overall sprint performance [10]. Specifically, faster individuals are able to produce higher pGRF, impulse and RFD during each foot strike when compared with slower individuals [10, 42]. Therefore, one possible explanation for our findings might be that, by increasing their lower-body strength levels, the subjects might have been able to produce higher pGRF, impulse and RFD after the training intervention, resulting in a greater running speed.

The present data also indicate that the experimental groups (lower-body resistance-training intervention) display a statistically significantly greater decrease ($p < 0.001$) in sprint time (mean $\pm$ SD = -3.11 ± 2.27 %; ES = -0.87) in comparison to the control groups (mean $\pm$ SD = -0.05 ± 2.13 %; ES = 0.02). Therefore, the reported reduction in sprint time resulting from resistance training (especially for elite and international athletes: mean $\pm$ SD = -4.07 ± 2.02 % and -2.34 ± 0.83 %, respectively) is likely to be worthwhile for athletes requiring high levels of speed as, according to the recommendation of

Fig. 3 Forest plot of sprint effect size ($n = 85$). Each *point* represents a sprint effect size and 95 % confidence interval. *CMJ* countermovement jump, *JS* jump squat, *plyo* plyometric, # standard deviation not reported



Hopkins [43], coaches and sport scientists should focus on enhancements as little as 0.3–1.5 % for elite athletes.

4.2 Effect of Subject Characteristics on Sprint Improvement

With respect to the subject characteristics, there is a moderate statistically significant ($r = 0.35$; $p = 0.011$)

correlation between body mass and sprint ES. Conversely, no statistically significant correlation was found between height ($r = 0.26$; $p = 0.08$), as well as age ($r = 0.03$; $p = 0.86$), and sprint ES (Table 1). In the present study, athletes' ages ranged from 13 to 25 years. It would be interesting to determine whether athletes over 25 years old can still experience transfer of lower-body strength training to sprint performance since athletes who require a high

level of speed, such as American footballers, basketballers, rugby and soccer players, usually reach their highest sporting performance between 22 and 26/28 years of age [44].

The results also indicate that improvement in sprint performance is dependent on the level of practice ($p = 0.03$) of the subjects, with national athletes exhibiting a greater sprint ES (ES = -1.24) than international (ES = -0.53), regional (ES = -0.31) and other (i.e., practicing below regional level) athletes (ES = -0.67). However, the large difference in the number of ESs available between national ($n = 22$), international ($n = 5$), regional ($n = 4$) and other ($n = 19$) athletes might explain this finding. Nevertheless, it is worth noting that when levels of practice are matched according to the number of ESs, improvement in sprint performance is greater with increasing level of practice.

4.3 Effect of Resistance-Training Program Elements on Sprint Improvement

With respect to the resistance-training program elements, the current meta-analysis shows that the improvement in sprint performance is independent ($p = 0.06$) of the resistance-training method used during the training intervention (Table 2). However, as only three ESs were included in the loaded JS/CMJ training method, versus 36 for the back squat training method and 13 for the back squat combined with loaded JS/CMJ and plyometric training method, the large difference in ESs available between the three resistance-training methods must be taken into consideration when interpreting these results. Additionally, the difference between the three resistance-training methods almost reached statistical significance ($p = 0.06$). The lack of statistical difference between the different resistance-training methods is in agreement with the findings of de

Villarreal et al. [45], who showed, in a recent meta-analysis, that different plyometric training methods induced a similar improvement in sprint performance. It is, however, worth noting that the back squat combined with loaded JS/CMJ and plyometric training method displays a greater sprint ES (ES = -1.20) than the back squat (ES = -0.81) and loaded JS/CMJ (ES = -0.29) training methods. Thus, from a practical standpoint, a mixed-method resistance-training approach (i.e., complex training) as recommended by Haff and Nimphius [46] appears to be the optimal training strategy for improving sprint performance when compared with more traditional training methods (resistance training or plyometric training alone). This is in line with previous research reporting a greater improvement in athletic performance after a mixed-method resistance-training intervention when compared with traditional training protocols [15, 47].

The improvement in sprint performance is also independent ($p = 0.34$) of the average load intensity of the resistance-training sessions (i.e., the average % 1 RM used throughout each training intervention) (Table 2). However, it is noteworthy that a lesser but non-statistically significant different sprint ES was found among studies using an average light intensity (i.e., 40–59.9 % of 1 RM; ES = -0.16) in comparison to studies using an average medium (i.e., 60–84.9 % of 1 RM; ES = -0.97), high (i.e., >85 % of 1 RM; ES = -0.52) and combination of high + very light (i.e., very light = <40 % 1 RM; ES = -0.82) intensity. This finding appears logical since it is generally accepted that an intensity >50 % 1 RM is necessary to induce gain in muscular strength [48] through peripheral (i.e., increase in muscle hypertrophy) or central (i.e., alterations in motor unit recruitment, increase in motor unit firing frequency, in motor unit synchronization, in motor unit excitability and in efferent drive to the muscle, and decrease in neural inhibition) adaptations [44]. Hence,

Table 2 Analysis for independent variables of resistance-training program elements

Program exercises						
Independent variables	Sprint improvement (%), mean $\pm$ SD	<i>F</i>	<i>p</i>	ES	SD	<i>n</i>
Type of exercise		$F(2,49) = 2.99$	$=0.06$			
Back squat	3.33 ± 2.33			-0.81	0.49	36
Back squat + loaded JS/CMJ + plyometric	3.20 ± 1.77			-1.20	1.34	13
Loaded JS/CMJ	0.02 ± 1.26			-0.29	0.25	3
Load intensity		$F(3,48) = 1.15$	$=0.34$			
High	5.02 ± 4.60			-0.52	0.23	3
Medium	3.26 ± 2.10			-0.97	0.92	34
Light	0.82 ± 0.60			-0.16	0.15	3
High + very light	2.78 ± 2.03			-0.82	0.55	12

CMJ countermovement jump, ES effect size, *F* Fisher–Snedecor distribution, JS jump squat, *n* sample size, *p* *p* value, SD standard deviation

given the strong correlation between back squat strength ES and sprint ES found in the present meta-analysis, medium, high and a combination of high and very light training load intensities were expected to induce a greater improvement in sprint performance (through an increase in strength levels) in comparison to light load training intensity. The large difference in ESs available in the present meta-analysis may explain the lack of statistical difference between the different training load intensities (Table 2). It is worth noting, however, that average high-intensity training resulted in lesser sprint ES ($ES = -0.52$) than medium ($ES = -0.97$) and a combination of high + light ($ES = -0.82$) training intensities. This result might be explained by the fact that resistance-training programs using an average high intensity (i.e., $>85\%$ of 1 RM) might have induced a greater stress (i.e., overwork), resulting in a smaller improvement in sprint performance. It is clear from the scientific literature that an athlete's ability to adapt to the training stimuli is reduced when high training intensities are sustained for too long [49]. From a practical perspective, using medium (i.e., $60\text{--}84.9\%$ of 1 RM) and a combination of high + very light (i.e., very light $<40\%$ 1 RM) training intensities appears to be an optimal training strategy for improving sprint performance.

The current meta-analysis indicates that there is a statistically significant correlation between the frequency of training sessions per week ($r = 0.50$; $p = 0.001$), as well as the rest interval between sets ($r = -0.47$; $p \leq 0.001$), and the magnitude of sprint improvement (Table 3). The positive correlation between the frequency of training sessions per week and the magnitude of sprint improvement indicates that higher frequencies of training resulted in lesser decreases in sprint time (i.e., positive ESs). As mentioned above, resistance-training programs including more than 2 sessions per week might have induced a greater stress (i.e., overwork), consequently resulting in a smaller improvement in sprint performance. The negative correlation between inter-sets rest interval of resistance

exercises and the magnitude of sprint improvement indicates that longer rest intervals resulted in greater decreases in sprint time (i.e., negative ESs). This result might be explained by the fact that longer rest intervals induced greater strength adaptations (in the present meta analysis, the correlation between inter-sets rest interval and increase in back squat strength was 0.43 , $p = 0.03$) resulting in greater decreases in sprint time. The greater strength adaptations with longer rest intervals is supported by Robinson et al. [50], who demonstrated that 2–3 min of rest between sets of resistance exercises resulted in greater increases in strength compared with shorter rest intervals (i.e., 30–90 s).

4.4 Effect of Resistance Training on Various Distances of Sprinting

The present data show no statistically significant differences ($p = 0.24$) among the different sprint distances (Table 4). This result might be explained by the fact that 96% (50 out of 52) of the sprint tests were less than 30 m. It is generally accepted that performance during short/medium sprints (<30 m) is highly dependent on 'speed strength' [51] and maximal power production [52], whereas performance during longer sprints (>30 m) depends on other factors, such as step frequency [53]. Accordingly, an increase in lower-body strength might provide a large benefit to performance in both short and medium sprints distances. Conversely, further research needs to be conducted in order to determine the impact of increasing lower-body strength on sprint performance over distances between 30 and 200 m.

5 Conclusion

The present meta-analysis suggests that there is a transfer of lower-body strength training to sprint performance as indicated by the very large correlation between squat strength ES and sprint ES ($r = -0.77$; $p < 0.001$). These data also indicate that an athlete's level of practice as well as the frequency of resistance-training sessions and the rest interval between sets of resistance exercises affect the magnitude of sprint performance improvement. Conversely, an athlete's age, resistance-training method, average training intensity ($\% 1$ RM), resistance-training duration, number of resistance-training exercises per session, number of sets per exercise and the number of repetitions per set do not appear to influence the magnitude of sprint performance improvement. Nevertheless, the large difference in the number of ESs available may account for these results and should be considered when interpreting these findings. From a practical standpoint, a mixed-

Table 3 Pearson's correlation coefficients (r) between various resistance-training program elements and sprint ES

Training program variables			
Independent variables	n	r	p
Frequency session/week	52	0.50	0.001
Program duration	52	-0.20	0.16
Number of exercises/session ^a	52	-0.20	0.16
Number of sets/exercise ^a	58	-0.27	0.06
Number of repetitions/sets ^a	52	-0.10	0.48
Rest between sets	50	-0.47	0.0006

ES effect size, n sample size, p p value

^a Average throughout the training intervention

Table 4 Analysis for independent variables of outcome measurement

Outcome measurement						
Independent variables	Sprint improvement (%), mean $\pm$ SD	<i>F</i>	<i>p</i>	ES	SD	<i>n</i>
Sprint test		$F(1,50) = 1.42$	$= 0.24$			
<20 m	3.60 ± 2.35			-0.99	1.01	27
20–40 m	2.58 ± 2.09			-0.73	0.52	25

ES effect size, *F* Fisher–Snedecor distribution, *n* sample size, *p* *p* value, SD standard deviation

method resistance-training approach (i.e., complex training) might be an optimal training strategy for improving sprint performance, rather than traditional training methods (resistance training or plyometric training alone). Additionally, resistance-training programs including 2 sessions per week of medium or combination of heavy + light training intensities (% 1 RM) might result in a greater improvement in sprint performance than those including 3 sessions per week and high training intensity.

Overall, the reported improvement in sprint performance (ES = -0.87, mean sprint improvement = 3.11 %) resulting from resistance training is of practical relevance for coaches and athletes in sport activities requiring a high level of speed, especially over short/medium distances (<30 m). To gain a greater understanding on the training variables affecting sprint performance, future meta-analyses should analyze variables of similar sample sizes (and therefore similar number of ESs). Future research should also equate the training volume of the resistance-training methods when comparing the effects of different resistance-training methods on sprint performance.

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